

Project Synopsis/Project Concept Document

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Project Title	Personal Health Chatbot
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Description

This project involves developing a "clinically aware" chatbot designed to answer queries based on a user's specific medical context rather than generic information. Standard models often lack personalization; for example, treatment advice must differ for a diabetic patient versus a non-diabetic one.

The proposed system synthesizes data from the client's existing EMR system (currently in structured JSON format) to create a comprehensive "Medical ID". This allows the chatbot to diagnose issues and provide advice that considers the patient's full history and uploaded reports. To adhere to strict privacy constraints, the architecture prioritizes PII protection, beginning with anonymized API models and migrating toward a fully self-hosted, local LLM.

Profile of Users

1. The Patient

- Role: The primary user seeking advice, uploading reports, and creating a medical ID.
- Usage: They use the system to interpret health issues in the context of their personal history.
- Characteristics: They require personalized diagnostics that account for chronic conditions and need assurance that their PII is anonymized and not exposed to public models.

2. The Doctor

- Role: A healthcare provider who utilizes the chatbot as a clinical decision support tool and an interface for rapid EMR retrieval.
- Usage: They use the system to perform rapid queries for longitudinal data (e.g., historical lab trends), generate concise summaries of complex patient histories, and explore diagnostic options in line with the patient's medical history.
- Characteristics: They require high-precision, hallucination-free insights that integrate with the existing EMR to reduce manual data review time.

3. The Administrator

- Role: Oversees the integration of EMR data and system architecture.
- Usage: They provide the structured medical data (JSON) and define the roadmap for deployment.
- Characteristics: Focused on data privacy and the technical transition from cloud-based APIs to an in-house LLM infrastructure.